Problem 4: ArrayList vs LinkedList

Random insert/delete

For random insertions and deletions in the middle of the list, `ArrayList` and `LinkedList` both have a time complexity of $O(n)$. However, in the results we observe that `ArrayList` is faster than `LinkedList`.

Sequential insert/delete

For sequential insertions and deletions, `LinkedList` is faster because it keeps head and tail references. Therefore `addFirst`, `addLast`, `removeFirst` and `removeLast` are all $O(1)$. For `ArrayList`, operations at the end are $O(1)$, but operations at the beginning are $O(n)$.

Random access

For random access, `ArrayList` is faster, since `get(i)` works in $O(1)$. In contrast, `LinkedList` has to traverse the nodes to find the element, which takes $O(n)$.